

PORT MACQUARIE, Australia

WMO: 947860

Lat: 31.4336S

Lon: 152.8656E

Elev: 5

StdP: 101.27

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	4.7	-1.3	3.4	12.7	0.1	3.8	11.3	9.9	14.4	8.9	15.2	1.1	300	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	30.5	22.9	29.1	22.5	28.0	22.2	24.5	28.5	23.7	27.3	23.1	26.5	5.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.0	26.5	22.6	17.3	25.9	22.0	16.7	25.2	74.1	28.7	71.2	27.4	68.7	26.6	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.4	0.1	36.3	1.1	3.5	-0.7	38.8	-1.4	40.8	-2.0	42.8	-2.8	45.3		
(5)			-0.3	25.7	1.0	0.8	-1.0	26.3	-1.6	26.8	-2.1	27.2	-2.9	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	22.9	22.7	21.5	18.8	15.8	13.7	12.6	13.2	15.6	17.7	19.7	21.6	(6)
(7)		DBStd	4.18	2.07	1.92	1.72	1.72	2.07	1.98	1.84	2.07	2.37	2.48	2.51	2.28	(7)
(8)		HDD10.0	5	0	0	0	0	0	1	2	2	0	0	0	0	(8)
(9)		HDD18.3	719	0	0	1	14	82	138	179	160	88	43	13	2	(9)
(10)		CDD10.0	2908	401	356	356	264	180	113	81	101	168	238	292	358	(10)
(11)		CDD18.3	579	143	122	98	28	3	0	0	1	6	22	54	102	(11)
(12)		CDH23.3	3357	934	726	448	113	12	1	2	11	72	155	297	587	(12)
(13)	CDH26.7	620	198	141	56	7	0	0	0	2	16	28	58	114	(13)	
(14)	Wind	WSAvg	3.3	3.8	3.7	3.3	2.9	2.8	2.9	2.8	3.0	3.3	3.6	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	1522	153	161	192	169	145	143	69	61	59	102	120	129	(15)
(16)		PrecMax	2582	551	354	678	619	382	473	287	259	198	396	282	373	(16)
(17)		PrecMin	798	28	28	10	7	7	7	1	0	0	13	13	17	(17)
(18)		PrecStd	431	116	81	135	144	98	99	58	59	44	82	65	81	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.5	30.1	27.6	24.7	22.2	22.9	24.7	29.1	29.6	31.0	32.3	(19)
MCWB			24.2	24.5	23.3	20.3	17.8	16.1	14.7	14.9	18.5	19.2	21.5	22.8	(20)	
2%		DB	30.1	29.5	28.0	25.8	23.1	20.7	20.4	21.9	25.2	26.7	28.2	29.2	(21)	
		MCWB	23.7	23.6	22.6	19.7	17.5	15.4	14.1	14.2	17.2	19.1	21.4	22.4	(22)	
5%		DB	28.6	28.2	26.9	24.6	21.9	19.5	18.9	20.4	23.2	24.9	26.3	27.7	(23)	
		MCWB	22.9	22.8	21.8	19.2	16.5	14.6	13.4	13.9	16.6	18.7	20.7	21.9	(24)	
10%		DB	27.5	27.1	25.8	23.5	20.6	18.4	17.7	19.1	21.6	23.3	24.8	26.4	(25)	
		MCWB	22.4	22.0	21.1	18.6	15.9	14.1	12.7	13.2	15.9	18.0	19.9	21.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.5	24.4	22.0	20.0	17.9	17.2	17.5	20.0	21.7	23.5	24.5	(27)
MCD B			30.2	30.0	28.4	25.3	22.7	20.6	19.8	21.5	25.1	26.1	28.6	28.9	(28)	
2%		WB	24.4	24.5	23.3	20.9	18.9	16.7	15.8	16.1	18.7	20.5	22.3	23.5	(29)	
		MCD B	28.5	28.2	26.5	24.3	21.5	19.1	18.1	19.5	22.9	24.5	26.3	27.5	(30)	
5%		WB	23.7	23.6	22.6	20.1	17.8	15.9	14.7	15.1	17.9	19.6	21.4	22.7	(31)	
		MCD B	27.3	26.8	25.5	23.3	20.3	18.2	17.4	18.6	21.7	23.5	25.0	26.3	(32)	
10%		WB	23.1	22.8	21.9	19.3	16.9	15.0	13.7	14.2	17.0	18.8	20.6	21.9	(33)	
		MCD B	26.4	25.9	24.8	22.4	19.5	17.3	16.6	17.8	20.4	22.3	23.8	25.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.1	8.3	9.4	10.2	9.9	11.1	12.4	12.3	10.6	9.0	8.8	(35)
MCDBR			10.1	10.1	10.1	11.2	11.8	11.4	13.4	15.1	14.9	13.4	11.6	10.8	(36)	
MCWBR			4.6	4.8	5.2	6.0	6.4	6.8	7.8	8.3	8.1	7.0	5.7	5.1	(37)	
5% WB		MCD BR	8.7	9.1	8.8	10.0	9.4	9.1	10.4	12.3	12.7	11.2	9.7	9.4	(38)	
		MCWBR	4.4	4.5	4.9	5.8	5.8	6.0	6.8	7.6	7.5	6.3	5.1	4.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.396	0.379	0.359	0.341	0.317	0.309	0.301	0.312	0.352	0.367	0.386	0.395	(40)
taud		2.472	2.542	2.593	2.607	2.631	2.642	2.651	2.603	2.486	2.480	2.472	2.466	(41)		
Ebn at Noon		944	941	928	893	870	853	880	912	919	942	947	948	(42)		
Edh at Noon		118	106	95	84	74	69	71	83	104	112	117	119	(43)		
(44)	All-Sky Solar Radiation	RadAvg		6.58	5.83	4.92	4.07	3.25	2.66	3.05	4.00	5.18	5.82	6.20	6.54	(44)
(45)		RadStd		0.47	0.46	0.41	0.27	0.23	0.19	0.25	0.26	0.32	0.48	0.59	0.53	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-60	+160	+86

Nomenclature: See separate page